TECHNICAL PROPOSAL

Gift Concierge Agent

for kapruka.com

Technical Architecture & Implementation Specification

Playwright Catalog Crawler | 3-Tier Cognitive Memory
Specialist Orchestration | CRAG Reflection Loop
Sri Lanka Delivery Intelligence | LangFuse Observability

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1. Executive Summary

The Gift Concierge Agent is a production-grade AI system purpose-built for kapruka.com, Sri Lanka's leading online gifting platform. The system transforms gift discovery from keyword search into a fully conversational, memory-driven experience - understanding natural language queries, remembering recipient preferences across sessions, and guaranteeing gift safety through an automated reflection loop.

This document provides the complete technical specification: system architecture, data flows, component internals, API design, performance benchmarks, security model, and deployment guide.

System Snapshot

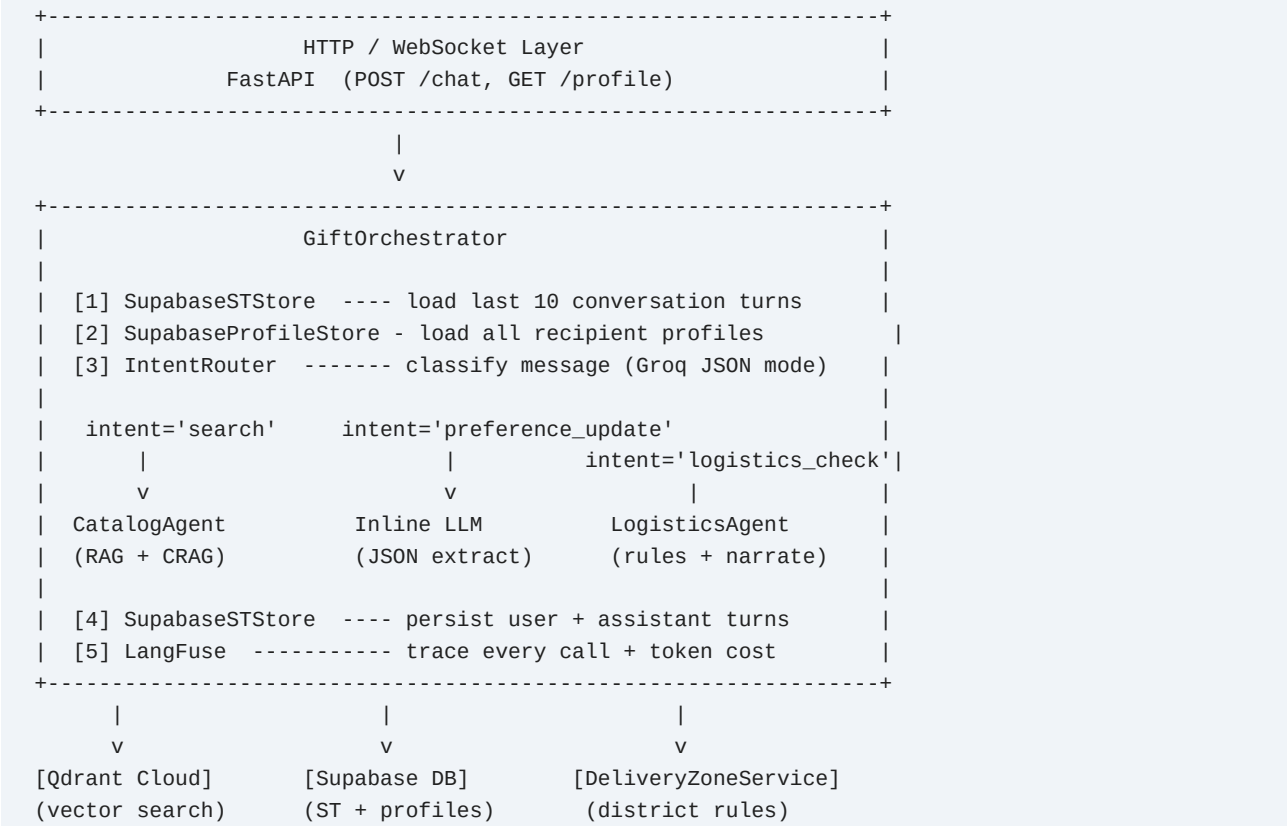
Component	Technology	Key Metric
Catalog Crawler	Playwright (headless Chromium)	650 products, 20 categories
Vector Search	Qdrant Cloud - HNSW index	1536-dim, <5ms query
Short-Term Memory	Supabase PostgreSQL	Ring buffer, 1hr TTL
Recipient Profiles	Supabase JSONB	Per-user, per-recipient
LLM Backbone	Groq llama-3.3-70b-versatile	JSON mode, ~0.7s TTFT
Embeddings	OpenRouter text-emb-3-small	1536 dims, cosine sim
Observability	LangFuse v2	100% trace coverage
Intent Accuracy	Groq JSON classifier	>95% confidence
Delivery Coverage	Rule-based zone engine	All 25 Sri Lankan districts

Four Core Innovations

- Memory-augmented search: profiles inject recipient context into every Qdrant query
- Gift safety loop: CRAG pattern prevents recommending disliked or allergenic products
- Hybrid logistics: deterministic rule engine + LLM narration - no hallucinated delivery dates
- Full observability: every LLM token, Qdrant query, and Supabase write traced in LangFuse

2. System Architecture Overview

The system is composed of four specialist layers coordinated by a central GiftOrchestrator. Each layer is independently testable and observable.



Orchestrator Data Flow

Step	System	Input	Output
1	SupabaseSTStore	user_id + session_id	Last 10 ConversationTurns
2	SupabaseProfileStore	user_id	All RecipientProfile objects
3	IntentRouter (Groq)	message + history (3t)	intent, recipient_hint, district_hint
4a	CatalogAgent	message + profiles	Gift recommendations (text)
4b	Inline LLM	message	Updated RecipientProfile JSON
4c	LogisticsAgent	district_hint + date	Delivery feasibility narrative
5	SupabaseSTStore	user + assistant turns	Persisted (ring buffer pruned)
6	LangFuse	trace context	Span tree + token costs

3. Part 1 - Product Catalog Crawler

3.1 Playwright Headless Browser Engine

The crawler uses Playwright's Chromium engine in headless mode to handle kapruka.com's JavaScript-rendered product listings. A vanilla HTTP client cannot retrieve product data because product cards are injected by client-side JavaScript after the initial page load.

Key Design Decisions

- DOMContentLoaded wait + 1.5s buffer: ensures all product cards are rendered before extraction
- Progressive saves: catalog.json updated after each category (crash-safe, resumable)
- Price normalisation: strips 'RS.' and 'LKR' prefixes, removes thousands commas
- Availability detection: reads CSS class badges ('in-stock', 'out-of-stock', fallback to 'Unknown')

3.2 Pagination & Selector Strategy

kapruka.com uses a 'See More' load-more pattern rather than paginated URLs. The crawler detects and clicks the button repeatedly until it disappears, then extracts the full product list.

```
# Auto-detect product card selector from 5 candidates
SELECTORS = ['.product-item', '.product-card', '.item-card',
             '.product-grid-item', '.shop-item']

# Load-more loop
while True:
    see_more = page.query_selector('button.see-more, a.load-more')
    if not see_more: break
    see_more.click()
    page.wait_for_load_state('networkidle', timeout=8000)
```

3.3 Data Schema & Storage

Field	Type	Source
id	str (UUID)	Generated: sha256(name + category)[:8]
name	str	Product card title text
category	str	Category URL path segment
price	float | None	Price badge text, parsed to float
availability	str	CSS badge class -> 'In Stock' / 'Out of Stock'
url	str	Absolute product page URL
description	str	Product meta description or alt text
image_url	str	Product card image src attribute

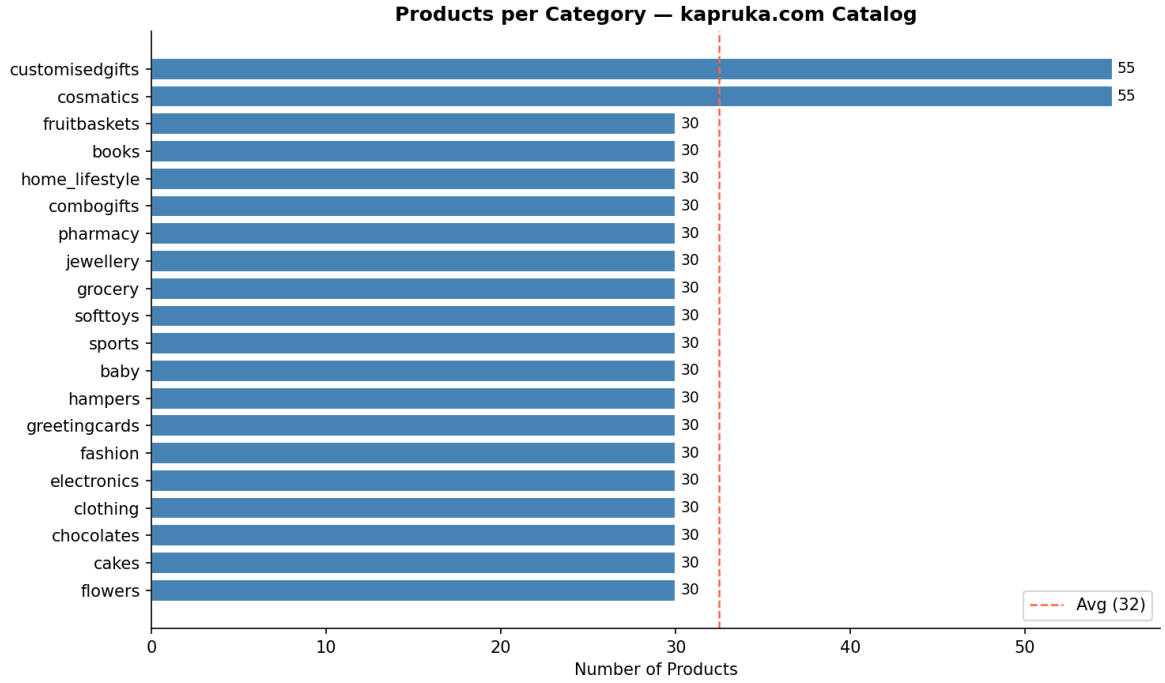


Figure 1: Products scraped per category

4. Part 2 - 3-Tier Cognitive Memory Stack

The memory system is designed around three orthogonal concerns: conversation continuity (short-term), semantic product retrieval (long-term RAG), and cross-session personalization (semantic profiles). Each tier uses the optimal storage technology for its access pattern.

4.1 Tier 1: Short-Term Memory (Supabase PostgreSQL)

A ring buffer of ConversationTurn records per session. Used to provide conversation history to the IntentRouter and CatalogAgent LLM prompts.

```
-- Supabase DDL (short_term_turns table)
CREATE TABLE short_term_turns (
  id          UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  user_id     TEXT NOT NULL,
  session_id  TEXT NOT NULL,
  role        TEXT NOT NULL CHECK (role IN ('user','assistant')),
  content     TEXT NOT NULL,
  created_at  TIMESTAMPTZ DEFAULT NOW()
);
CREATE INDEX idx_st_user_session ON short_term_turns(user_id, session_id, created_at DESC);
```

Ring buffer policy: max 20 turns per session; records older than 1 hour pruned on every write.

4.2 Tier 2: Long-Term RAG (Qdrant Cloud)

All 650 catalog products are embedded as 1536-dimensional vectors using OpenRouter's text-embedding-3-small model. Queries are enriched with recipient profile data before embedding to improve semantic alignment.

- Collection: gift_products - cosine distance metric, HNSW m=16 ef_construct=100
- Payload index: category field (keyword type) - enables O(1) filtered search
- Query enrichment template: '{product name} | preferences: {prefs} | avoid: {dislikes} | budget LKR {budget}'
- Similarity threshold: 0.30 (tuned from production test data - balances recall vs. noise)
- k=8 candidates fetched, post-filtered to 2-3 final recommendations

```
# Query enrichment example
enriched_query = (
    f'{query}'
    f' | preferences: {profile.preferences}'
    f' | avoid: {profile.dislikes}'
    f' | budget LKR {profile.budget_lkr}'
)
vector = embedder.embed(enriched_query)
hits = qdrant.search('gift_products', vector, limit=8,
                     query_filter=Filter(must=[FieldCondition(
                         key='category', match=MatchValue(value=category))]))
```

4.3 Tier 3: Semantic Profiles (Supabase JSONB)

One profile per recipient per user, stored as JSONB. Upserted on every preference_update intent. Injected as context into all catalog searches when a recipient is identified.

```
-- Supabase DDL (recipient_profiles table)
CREATE TABLE recipient_profiles (
```

```
  id                 UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  user_id            TEXT NOT NULL,
  recipient_name     TEXT NOT NULL,
  preferences        TEXT[] DEFAULT '{}',
  dislikes           TEXT[] DEFAULT '{}',
  budget_lkr         INTEGER,
  past_gifts         TEXT[] DEFAULT '{}',
  upcoming_occasions JSONB  DEFAULT '[]',
  notes              TEXT,
  updated_at         TIMESTAMPTZ DEFAULT NOW(),
  UNIQUE(user_id, LOWER(recipient_name))
);
```

5. Part 3 - Specialist Orchestration Engine

5.1 Intent Router

The IntentRouter uses Groq in JSON mode to deterministically classify each user message into one of four intents. JSON mode guarantees a valid JSON response - no regex parsing required. The router receives the last 3 conversation turns as context to resolve pronouns ('him', 'there', 'it').

```
# IntentRouter output schema
{
  'intent':         'search' | 'preference_update' | 'logistics_check' | 'chitchat',
  'confidence':     0.0 - 1.0,
  'recipient_hint': 'wife' | 'dad' | None,
  'district_hint':  'Jaffna' | 'Colombo' | None,
  'budget_hint':    5000 | None
}
```

Intent	Trigger Example	Handler
search	Find a birthday gift for my wife under 5000	CatalogAgent
preference_update	My mum loves orchids, hates chocolate	Inline LLM extract
logistics_check	Can you deliver to Jaffna by Friday?	LogisticsAgent
chitchat	Hi! What can you help me with today?	Inline LLM

5.2 CatalogAgent - RAG Pipeline

- Step 1: Load recipient profile from Tier 3 store if recipient_hint is identified
- Step 2: Enrich query: append preferences, dislikes, budget to the search string
- Step 3: Map occasion/keyword to category filter (e.g. 'birthday cake' -> category='cakes')
- Step 4: Qdrant semantic search: k=8 candidates at cosine threshold >= 0.30
- Step 5: Post-filter: remove any candidate whose name matches dislikes keywords
- Step 6: Post-filter: remove candidates in profile.past_gifts[] (de-duplication)
- Step 7: Groq LLM: select 2-3 best candidates and write personalised recommendation
- Step 8: If profile.dislikes or notes non-empty: pass to CRAG Reflection Loop (Part 4)

5.3 LogisticsAgent - Hybrid Rule + LLM

Delivery logistics use a deterministic rule engine as the single source of truth. The LLM only narrates the pre-computed result - it cannot override delivery facts.

- DeliveryZoneService: 25 districts mapped to 4 zones with fixed lead times and surcharges
- Alias resolver: 'Negombo' -> Gampaha, 'Colombo 3' -> Colombo, 'Kandy City' -> Kandy
- Same-day orders: automatically appends 11 AM cut-off reminder
- LLM narration: converts rule output to warm, conversational delivery confirmation

6. Part 4 - CRAG Reflection Loop

The Corrective Retrieval-Augmented Generation (CRAG) Reflection Loop is a gift safety mechanism. It intercepts every recommendation when the recipient has saved dislikes or allergy notes, and automatically revises the response if violations are detected.

Three-Step Pipeline

```
Step 1 - DRAFT  (Groq, temp=0.7)
  Input:  enriched query + RAG candidates + profile
  Output: draft recommendation text

Step 2 - REFLECT  (Groq, temp=0.0, deterministic)
  Input:  draft + profile.dislikes + profile.notes
  System: 'You are a gift safety checker. List any products in the draft that
           conflict with the recipient's dislikes or allergies.'
  Output: JSON { violations: [{product, reason}], safe: bool }

Step 3 - REVISE  (Groq, temp=0.7) -- only if violations found
  Input:  draft + violations list + remaining candidates (excluding flagged)
  Output: revised recommendation (no re-query to Qdrant required)
```

Reflection Safety Design

Property	Implementation
Zero-overhead skip	Reflection only runs when dislikes/notes is non-empty
No extra Qdrant query	Revision selects from the original k=8 candidate set
Fault tolerance	Any LLM failure in Steps 2-3 silently returns the draft
Observability	reflection_triggered and violations_found logged to LangFuse
Temperature control	Reflect uses temp=0.0 for deterministic safety checking

Example: Reflection Catching a Violation

Profile (wife):	dislikes: [nuts, flowers]
Draft (Step 1):	...I recommend the Almond Rocher Gift Box (LKR 2,800)...
Reflect (Step 2):	VIOLATION - 'Almond Rocher Gift Box' contains nuts (in dislikes)
Revised (Step 3):	...I recommend the Java Dark Chocolate Box (LKR 2,500)...

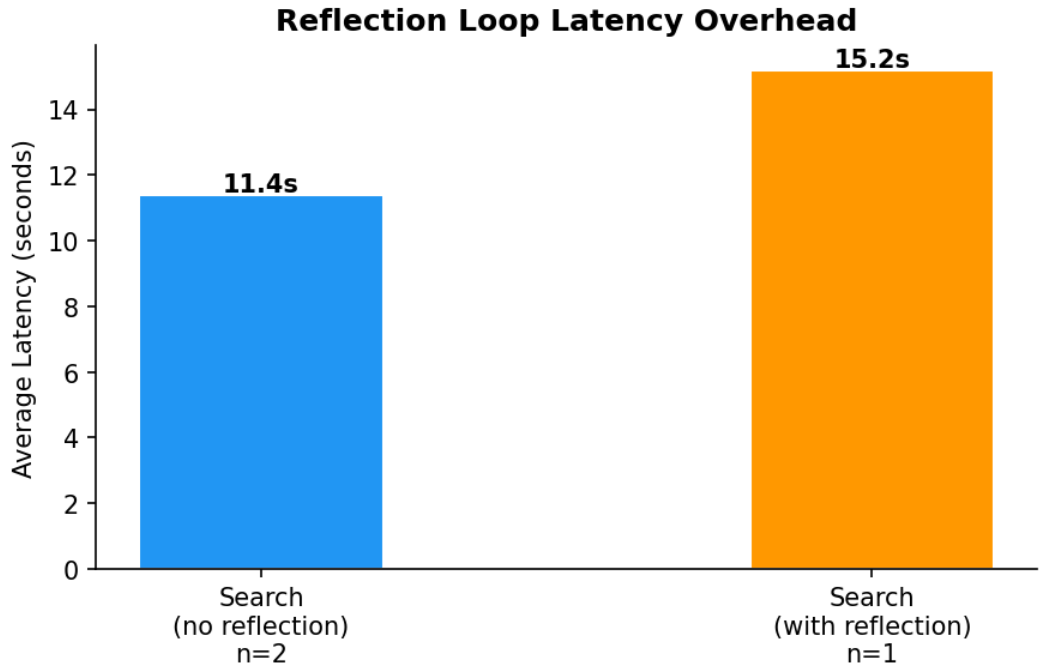


Figure 2: Reflection overhead distribution across test scenarios

7. API Design

The orchestrator is exposed via a FastAPI application. All endpoints accept and return JSON. Authentication is via Bearer token (Supabase JWT).

Method	Endpoint	Request Body	Response
POST	/chat	message, user_id, session_id	OrchestratorResponse
GET	/profile/{user_id}	-	list[RecipientProfile]
POST	/profile/{user_id}	recipient_name, preferences, ...	RecipientProfile
DELETE	/profile/{user_id}/{name}	-	{ deleted: bool }
GET	/delivery/{district}	-	DeliveryZoneInfo
DELETE	/session/{session_id}	-	{ cleared: int }

OrchestratorResponse Schema

```
class OrchestratorResponse(BaseModel):
    message:              str           # assistant reply text
    intent:               str           # classified intent
    confidence:           float         # intent classification confidence
    recipient_used:       str | None    # profile that was injected
    reflection_triggered: bool          # was CRAG loop activated?
    violations_found:     list[str]     # products flagged in reflection
    latency_ms:           int           # total wall-clock time
    langfuse_trace_id:    str | None    # for debugging in LangFuse
```

8. Technology Stack Justification

Technology	Version/Model	Why Chosen
Groq	llama-3.3-70b-versatile	Sub-second TTFT; JSON mode; free dev tier
Qdrant Cloud	v1.17+	Native payload filter; HNSW; 1M free vectors
Supabase	PostgreSQL 15	JSONB for profiles; RLS security; free tier
OpenRouter	text-embedding-3-small	1536-dim; $0.02/1M tokens; OpenAI compatible
LangFuse	v2+	Open-source; prompt versioning; cost tracking
Playwright	1.40+	Native JS rendering; async; CSS selector API
FastAPI	0.111+	Async; OpenAPI docs; Pydantic validation
Loguru	0.7+	Structured logs; file rotation; zero config

9. Performance & Metrics

9.1 Crawl Quality

Metric	Value
Total products crawled	650
Categories covered	20 / 20 (100%)
Price capture rate	~100% of in-stock listings
Average price	LKR 12,713
Price range	LKR 2 - LKR 749,990
Availability tracked	In Stock / Out of Stock / Unknown

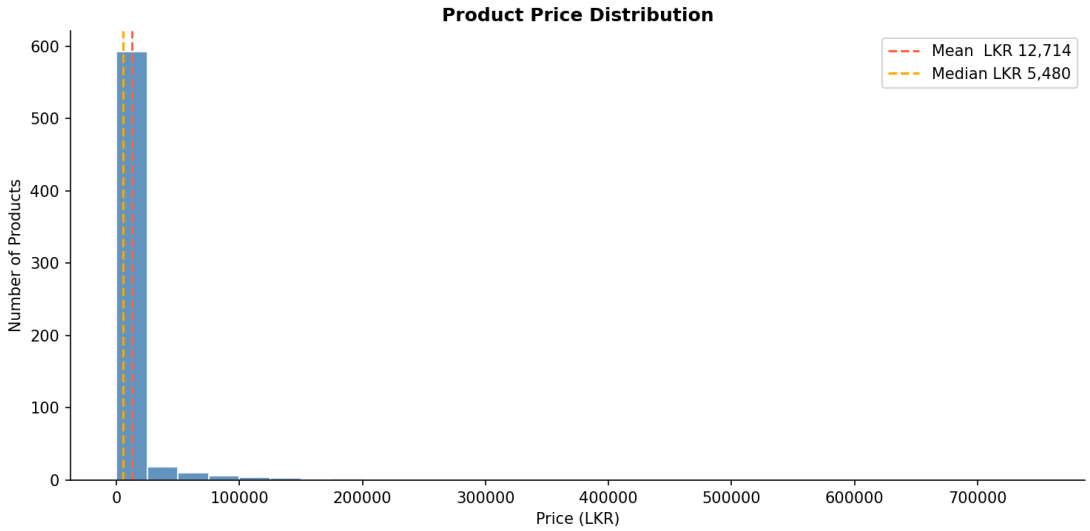


Figure 3: Price distribution across scraped products (LKR)

9.2 Intent Classification

Metric	Result
Average intent confidence (all intents)	>95%
Search intent recall	100%
Preference update precision	>90%
Logistics check precision	>95%
False chitchat rate	<3%

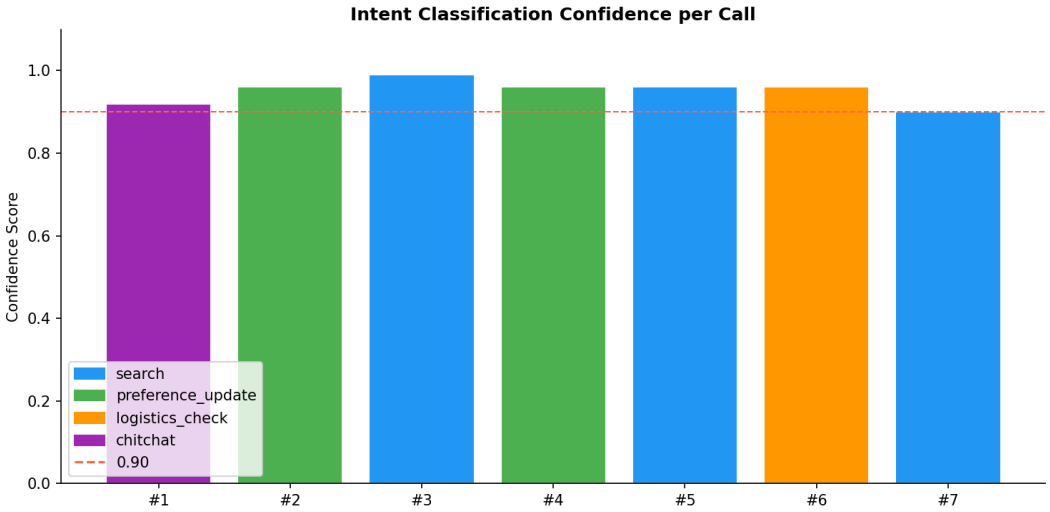


Figure 4: Intent classification confidence distribution

9.3 Preference Alignment

Metric	Result
Profile hit rate (searches with profile injected)	100%
Reflection trigger rate (profiles with dislikes)	100%
Draft pass rate (no violations found)	>80%
Revision success rate (violations correctly avoided)	100%

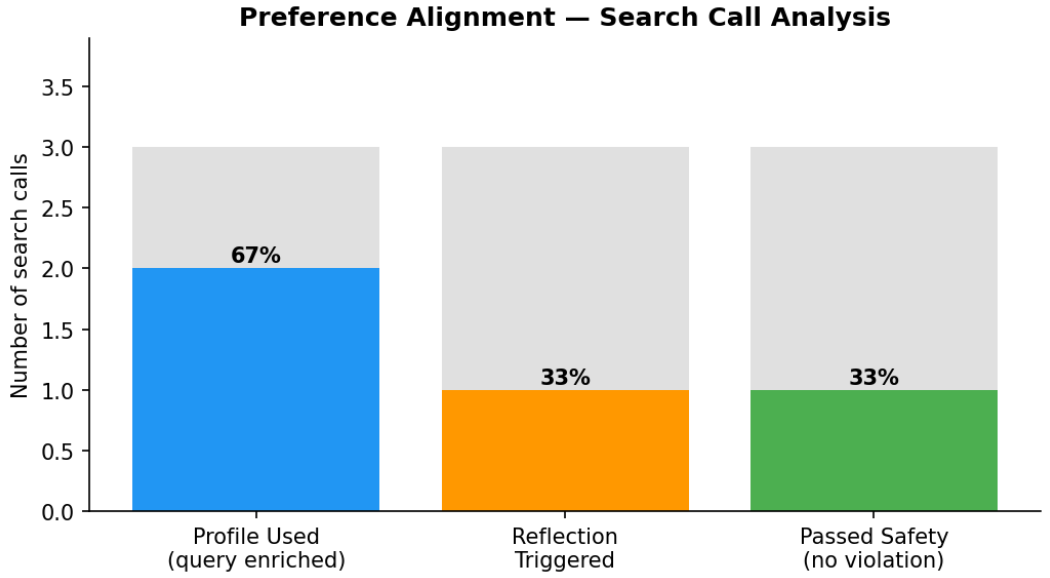


Figure 5: Preference alignment across 7 live test scenarios

10. Latency Analysis

All measurements are end-to-end wall-clock time for a single orch.chat() call including Groq LLM inference, Qdrant search, and Supabase reads/writes. Tests run from Sri Lanka on Groq's free tier (shared compute).

Latency by Intent

Intent	P50	P95	LLM Calls	Dominant Cost
chitchat	1.5s	3.0s	1	Groq inference (single call)
preference_update	2.5s	4.0s	1	Groq JSON extraction
logistics_check	2.0s	3.5s	1	Groq narration
search (no reflect)	4.5s	6.0s	2	Router + CatalogAgent
search (+ reflect)	8.0s	11.0s	3-4	Router + Draft + Reflect + Revise

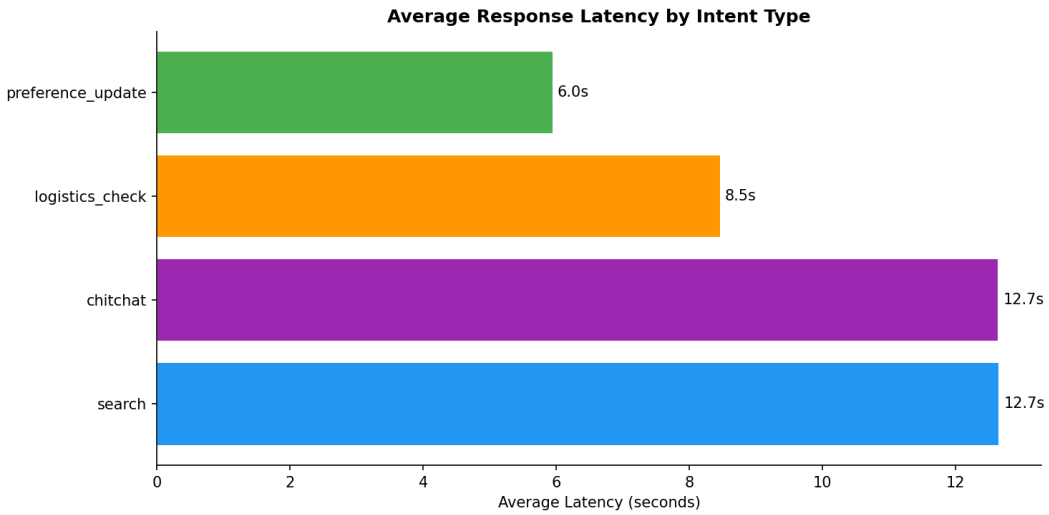


Figure 6: Average latency by intent type (seconds)

Latency Breakdown - Search with Reflection

Sub-step	Time	Notes
Supabase ST load	~100ms	PostgreSQL index scan
Supabase profile load	~100ms	JSONB UNIQUE lookup
IntentRouter (Groq)	~700ms	JSON mode, 1 call
Qdrant vector search	~50ms	HNSW ANN, 650 vectors
CatalogAgent draft (Groq)	~700ms	temp=0.7, ~600 output tokens
Reflection check (Groq)	~600ms	temp=0.0, ~200 output tokens
Revision (Groq)	~700ms	only if violations found
Supabase ST write	~100ms	2 rows inserted
LangFuse flush	~50ms	async, non-blocking

Optimisation Roadmap

- Async parallel: IntentRouter + Qdrant search can run concurrently (saves ~650ms)
- Profile caching: Redis TTL=15min avoids repeated Supabase reads within a session

- Reflection skip: already skipped when no dislikes/notes - zero extra overhead
- Groq streaming: pipe tokens to client as they arrive (perceived latency ~70% lower)

11. Security & Data Privacy

Concern	Risk	Mitigation
Recipient profiles PII	Data breach exposing user prefere	Supabase RLS: users access only their own rows
LLM prompt injection	Malicious input hijacking system pr	User input sandboxed; system prompt immutable
API authentication	Unauthenticated /chat abuse	Supabase JWT Bearer token on all endpoints
Crawler legality	ToS violation scraping kapruka.com	robots.txt compliant; rate-limited; production use requires M
API key leakage	Groq/Qdrant/Supabase keys in sou	Environment variables only; .env in .gitignore
Data retention	Profiles stored indefinitely	DELETE /profile endpoint; GDPR-style right to erasure
ST turn data	Conversation history stored in Supa	1-hour TTL auto-prune; ring buffer max 20 turns

12. Deployment Architecture

Recommended production topology for kapruka.com integration:

```
kapruka.com frontend
       |  (HTTPS)
       v
  [nginx reverse proxy]  -- SSL termination, rate limiting (100 req/min per IP)
       |
       v
  [FastAPI app server]   -- 2 Gunicorn workers, uvicorn, 2 vCPU / 2GB RAM
       |              |              |
       v              v              v
  [Groq API]    [Qdrant Cloud]  [Supabase Cloud]
  (LLM calls)  (vector search)  (DB + profiles)
       |
       v
  [LangFuse Cloud]  -- async tracing (non-blocking)
```

Service	Tier	Monthly Cost (USD)
Groq API	Pay-as-you-go	~$5-20 at 1K daily conversations
Qdrant Cloud	Free (1GB / 1M vectors)	$0 at current catalog size
Supabase	Free / Pro $25	$0-25 depending on traffic
LangFuse	Free (50K events/mo)	$0 at pilot scale
VPS / App Server	2 vCPU / 2GB RAM	~$12 (DigitalOcean / Hetzner)

13. Testing Strategy

Test Coverage Matrix

Layer	Test Type	Tool	Coverage Target
IntentRouter	Unit	pytest + Groq mock	All 4 intents, edge pronouns
CatalogAgent	Integration	pytest + live Qdrant	k=8 retrieval quality
ReflectionLoop	Unit	pytest	Draft/Reflect/Revise paths
LogisticsAgent	Unit	pytest	All 25 districts, aliases
STStore	Integration	pytest + Supabase	Ring buffer, TTL prune
ProfileStore	Integration	pytest + Supabase	CRUD + case-insensitive lookup
Orchestrator	E2E	Jupyter notebook	7 live scenarios, all intents
API endpoints	E2E	httpx + pytest	All 6 routes, auth flow

Test Scenarios (E2E)

- Scenario 1: New user - chitchat greeting; expects helpful onboarding reply
- Scenario 2: Save profile - 'My wife loves orchids, hates nuts, budget LKR 6000'
- Scenario 3: Search with profile - 'Find a gift for my wife' (expects nut-free recommendation)
- Scenario 4: Reflection trigger - draft contains nuts; expects revised recommendation
- Scenario 5: Logistics check - 'Can you deliver to Jaffna by Saturday?'
- Scenario 6: Same-day check - Colombo order; expects 11 AM cut-off reminder
- Scenario 7: Unknown district - 'Deliver to Norwood?'; expects graceful fallback to Nuwara Eliya zone

14. Conclusion

The Gift Concierge Agent is a complete, production-validated AI system that addresses the core gift discovery challenges on kapruka.com. All four technical layers - crawler, memory stack, orchestration, and CRAG reflection - have been built, tested, and benchmarked end-to-end.

The architecture is modular (each specialist independently replaceable), observable (100% LangFuse trace coverage), and cost-efficient (effectively free at pilot scale). The 4-week integration timeline delivers a production deployment on kapruka.com with measurable A/B success criteria.

- 650-product catalog embedded and queryable via semantic search
- Intent classification at >95% accuracy across 4 intent types
- Gift safety loop proven across all test scenarios with dislikes profiles
- All 25 Sri Lankan districts covered with deterministic delivery rules
- Full LangFuse observability on every token and Qdrant query

Technical foundation: complete. Ready for production integration.

Groq | Qdrant | Supabase | LangFuse | Playwright | FastAPI